

Name:

Answers

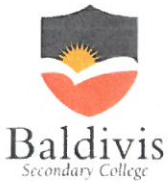
Date: _____

Teacher: _____

Year 9 2018

End of Term 1 Test

/ 58

**Full working out MUST be shown to get full marks for each question.**

Total Time: 55 minutes

Weighting: 9%

Equipment: To be provided by the student: Pen, pencil, ruler, scientific calculator, 1 - 2-sided page of notes

Question 1**4 marks**

Simplify the following leaving the answer in index notation.

a. $a^3 \times a^5$

a^8 ✓

b. $30a^6 \div 10a^4$

$3a^2$ ✓

c. $(5x^4)^3$

$125x^{12}$ ✓

(4)

Question 2**4 marks**

Evaluate the following:

a. $4x^0$

$4 \times 1 = 4$ ✓

b. 2^{-3}

$\frac{1}{2^3} = \frac{1}{8}$ ✓

(4)

Question 3**10 marks**

Use distributive law to expand these expressions:

a. $3(x+2)$

$3x + 6$ ✓

b. $3x(2x+y)$

$6x^2 + 3xy$ ✓

c. $7(4-x) + 9x$

$28 - 7x + 9x$ ✓
 $= 28 + 2x$ ✓

d. $3(x+4) + 5(x-7)$

$3x + 12 + 5x - 35$ ✓

$8x - 27$ ✓

e. $x(x+4) - 3(x-7)$

$x^2 + 4x - 3x + 21$ ✓

$x^2 + x + 21$ ✓

(10)

(18)

Expand these binomial Products

a. $(x+2)(x+5)$

$$x^2 + 7x + 10$$

(1) (1)

b. $(x+3)(x-7)$

$$x^2 - 4x - 21$$

(1) (1)

c. $(x-4)(x-7)$

$$x^2 - 11x + 28$$

(1) (1)

Question 5

5 marks

Find the Highest Common Factor of these expression

a. $xy + 4x$

$$(x)$$

(1)

b. $10xy + 5y$

$$(5y)$$

(1)

c. $16(x+2) + y(x+2)$

$$(x+2)$$

(1)

d. $(x+4)^2 + 5(x+4)$

$$(x+4)$$

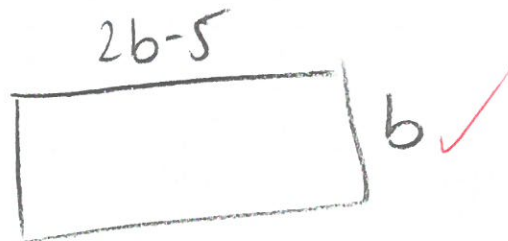
(2)

Question 6

5 marks

A photo frame has a side length of b . its top length is $2b - 5$.

a. Draw a picture of the photo frame.



(1)

b. Write an expression for the Perimeter of the picture frame.

$$2b - 5 + 2b - 5 + b + b =$$

$$6b - 10$$

(2)

c. Write an expression for the area of the picture frame.

$$2b^2 - 5b = b \times 2b - 5$$

Ans w.o

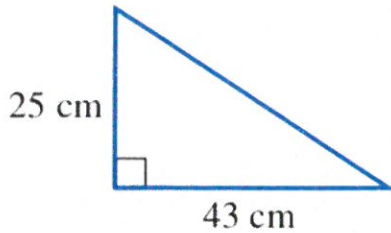
(2)

Question 7

8 marks

Find the area of these shapes.

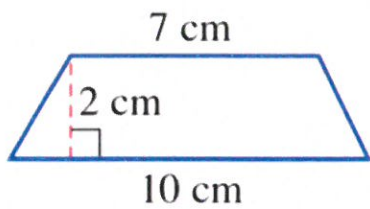
a.



$$\frac{25 \times 43}{2} = 537.5 \text{ cm}^2$$

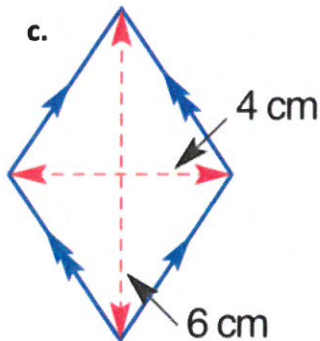
-1 overall
No units

b.



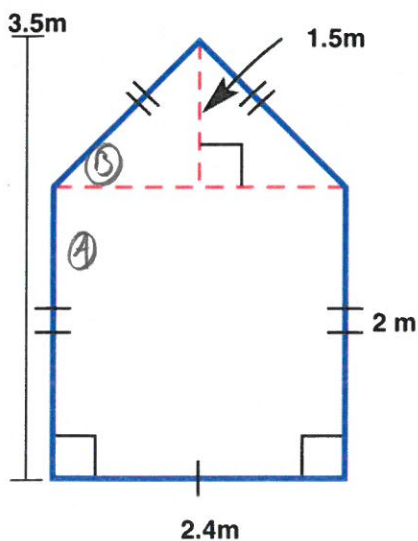
$$\left(\frac{10 + 7}{2} \right) \times 2 = 17 \text{ cm}^2$$

c.



$$\frac{4 \times 6}{2} = 12 \text{ cm}^2$$

d.



$$\begin{aligned} A &= 2 \times 2.4 = 4.8 \\ B &= \frac{1}{2} \times 2.4 \times 1.5 = 1.8 \\ \hline 4.8 + 1.8 &= 6.6 \text{ m}^2 \end{aligned}$$

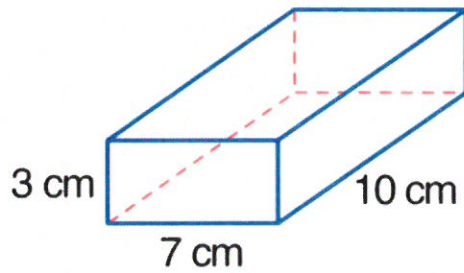
8

Question 8

4 marks

Find the volume of these shapes.

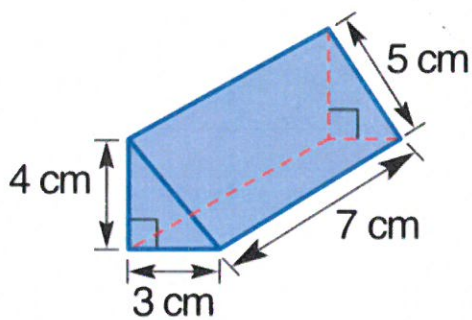
a.



$$\begin{aligned} & \underline{3 \times 7 \times 10} \\ & 21 \times 10 = \underline{210 \text{ cm}^3} \end{aligned}$$

✓ ✓

b.

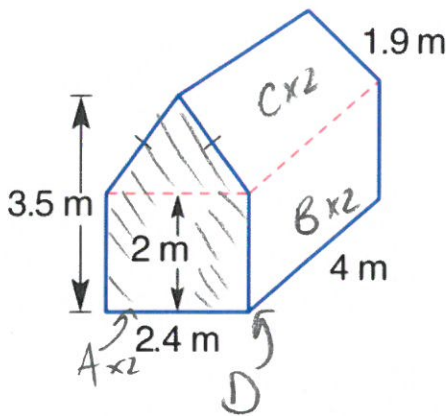


$$\frac{1}{2} \times 4 \times 3 \times 7 = \underline{42 \text{ cm}^3}$$

✓ ✓

Farmer Kath is building a new water tank and needs to know how much metal to order, and how much water it will hold.

- a. Find the total surface area of the design below.



Answer from

$$A = (7d) = 6.6 \text{ m}^2 \times 2 = 13.2 \text{ m}^2$$

$$B = 2 \times 4 \times 2 = 16 \text{ m}^2$$

$$C = 1.9 \times 4 \times 2 = 15.2 \text{ m}^2 \quad \checkmark$$

$$D = 2.4 \times 4 = 9.6 \text{ m}^2 \quad \checkmark$$

$$\underline{\underline{51.3 \text{ m}^2 \quad \checkmark}} \quad (2)$$

- b. Find the volume of the water tank.

$$13.2 \times 4 = \boxed{52.8 \text{ m}^3} \quad (2)$$

- c. Find the total capacity of the water tank, if $1 \text{ m}^3 = 1000 \text{ L}$

$$52.8 \times 1000 = \underline{\underline{52,800 \text{ L}}} \quad (2)$$

- d. Farmer Kath goes to the bank and gets a simple interest loan of \$10,500 to pay off the tank, at a rate of 6.2% for 5 years.

- i. What is the interest amount that farmer Kath has to pay?

$$10,500 \times 0.062 \times 5 = \underline{\underline{\$3255}} \quad (2)$$

- ii. What is the gross (total) amount that Farmer Kath has to pay off after 5 years?

$$3255 + 10500 = \underline{\underline{\$13,755}} \quad (2)$$

- iii. If Farmer Kath pays off the loan with \$50 a week, how much would she still owe after the 5-year period? (52 weeks in a year?)

$$50 \times 52 \times 5 = \underline{\underline{\$13,000}} - \$13,755 \quad (2)$$

$$\boxed{\$755 \text{ left}} \quad \checkmark$$

